EE5121: Convex Optimization

Assignment 2

Rohith R — EP21B030

Contents

Maximum Entropy with Moment Constraints **2**
- Dual derivation . . . 2
- Strong duality conditions . . . 2
- Gibbs form of the primal maximizer . . . 3
- Dual in terms of θ and role of $\theta^\star$ . . . 3
- Numerical primal solution . . . 3
- Dual solution and primal reconstruction . . . 4

Logistic Regression **5**
- Probabilistic interpretation . . . 5
- Numerical solver . . . 5
- Why no finite optimizer exists . . . 5
- Regularized formulation . . . 6

Graph Coloring SDP **8**
- Dual derivation . . . 8
- Numerical results . . . 8
- Feasibility of 3-colorability . . . 9

All scripts used in this assignment are available in the ZIP archive: `Assignment_2_Code.zip`.

Question 1: Maximum Entropy with Moment Constraints

We consider

$$\max_{p\in\Delta_n} H(p) = -\sum_{i=1}^{n} p_i \log p_i \quad \text{s.t.} \quad \sum_i p_i\phi_i = \mu,\ \sum_i p_i = 1.$$

(a) Dual derivation

The Lagrangian with multipliers $\theta \in \mathbb{R}^k$ (moment) and $\nu \in \mathbb{R}$ (simplex) is

$$\mathcal{L}(p,\theta,\nu) = -\sum_i p_i \log p_i + \theta^\top\Big(\sum_i p_i\phi_i - \mu\Big) + \nu\Big(\sum_i p_i - 1\Big).$$

Stationarity w.r.t. p_i gives

$$-(1+\log p_i) + \theta^\top\phi_i + \nu = 0 \ \Rightarrow\ p_i = \exp(\theta^\top\phi_i + \nu - 1).$$

Substituting into $\mathcal{L}$ yields the dual function:

$$\boxed{g(\theta,\nu) = e^{\nu-1}\sum_{i=1}^{n} e^{\theta^\top\phi_i} - \theta^\top\mu - \nu}.$$

The dual problem (in θ,ν) is

$$\min_{\theta\in\mathbb{R}^k,\ \nu\in\mathbb{R}} \ g(\theta,\nu)$$

(b) Strong duality conditions

The constraint set is affine, and the objective $H(p)$ is strictly concave on the relative interior of the simplex. Slater's condition requires the existence of a *strictly feasible* point, i.e.,

$$p_i > 0 \ \ \forall i, \qquad \text{and} \qquad \sum_{i=1}^{n} p_i\phi_i = \mu.$$

Such a distribution exists if and only if

$$\mu \in \operatorname{relint}\big(\operatorname{conv}\{\phi_1,\dots,\phi_n\}\big).$$

Geometrically, this means that μ must lie in the *strict interior* of the convex hull of the feature vectors, so that it admits a representation as a strictly convex combination of ϕ_i. Under this condition the feasible set has nonempty relative interior, and the dual optimum is attained with zero duality gap:

$$H(p^\star) = g(\theta^\star,\nu^\star).$$

Moreover, the primal solution is unique and takes the exponential-family form

$$p_i^\star \propto \exp\big(\theta^{\star\top}\phi_i\big).$$

(c) Gibbs form of the primal maximizer

From stationarity,

$$p_i^\star = \exp(\theta^{\star\top}\phi_i + \nu^\star - 1) = \frac{\exp(\theta^{\star\top}\phi_i)}{\sum_j \exp(\theta^{\star\top}\phi_j)} \quad \text{Proof in (d).}$$

Thus

$$\boxed{p_i^\star = \frac{e^{\theta^{\star\top}\phi_i}}{\sum_j e^{\theta^{\star\top}\phi_j}}}$$

and $\sum_i p_i^\star \phi_i = \mu$.

(d) Dual in terms of θ and role of $\theta^\star$

Maximizing $g(\theta, \nu)$ over ν gives

$$\frac{\partial g}{\partial \nu} = e^{\nu-1}\sum_i e^{\theta^\top\phi_i} - 1 = 0 \;\Rightarrow\; e^{\nu-1} = \frac{1}{\sum_i e^{\theta^\top\phi_i}}.$$

Plugging back yields the reduced dual

$$\boxed{g(\theta) = \log\Big(\sum_{i=1}^n e^{\theta^\top\phi_i}\Big) - \theta^\top\mu}$$

with gradient

$$\nabla g(\theta) = \sum_i \phi_i \frac{e^{\theta^\top\phi_i}}{\sum_j e^{\theta^\top\phi_j}} - \mu = \mathbb{E}_{p(\theta)}[\phi] - \mu.$$

Hence $\nabla g(\theta^\star) = 0$ is the moment-matching condition: $\theta^\star$ is the natural parameter of the maximum-entropy exponential family matching μ.

(e) Numerical primal solution

The primal problem is solved in CVXPY using the concave entropy atom `entr`. This yields the optimal distribution $p^\star$ with entropy

$$H(p^\star) = 2.270053595467961.$$

To study convergence, we generate iterates by performing gradient descent on the dual objective $g(\theta)$ and mapping each θ^k back to a primal iterate via the Gibbs map

$$p_i^k = \frac{\exp(\theta^{k\top}\phi_i)}{\sum_j \exp(\theta^{k\top}\phi_j)}.$$

For each k we record:

$$H(p^k), \qquad \|\Phi^\top p^k - \mu\|_2, \qquad \big|\mathbf{1}^\top p^k - 1\big|,$$

which illustrate the convergence of the iterates toward feasibility and optimality.

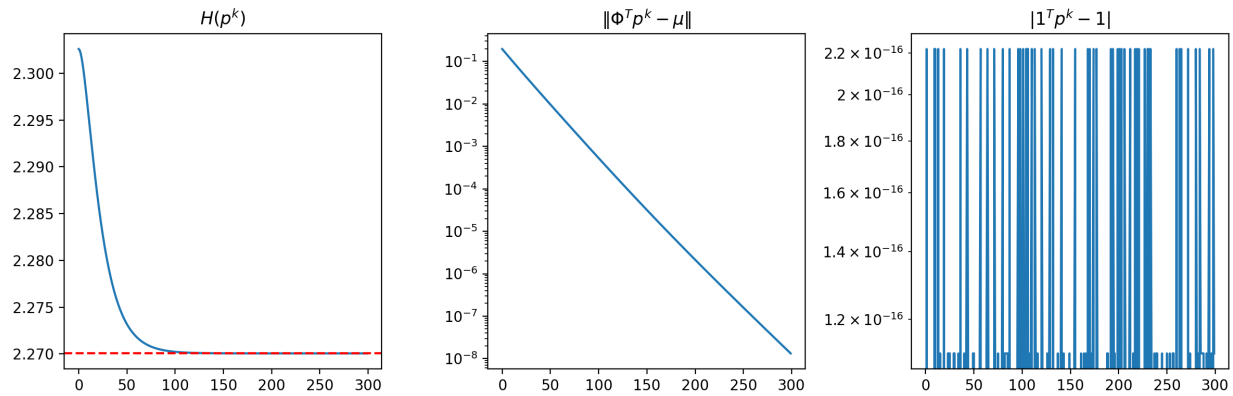


Figure 1: Iterate diagnostics for part (e): (left) entropy $H(p^k)$ with the optimal value marked, (middle) feasibility residual $\|\Phi^\top p^k - \mu\|_2$ (log scale), (right) simplex error $|\mathbf{1}^\top p^k - 1|$.

Remark. Let $r_k := \|\Phi^\top p_k - \mu\|_2$. For smooth updates on the dual (or mirror–descent updates on p_k), the dual objective $g(\theta)$ is locally α–strongly concave with L–Lipschitz gradient, which yields the standard linear contraction

$$r_{k+1} = \|\nabla g(\theta_{k+1})\| \;\le\; \left(1 - \frac{\alpha}{L}\right) r_k = \rho\, r_k, \qquad 0 < \rho < 1.$$

Thus

$$\log r_k = k \log \rho + \text{const},$$

so the feasibility residual appears as a straight line when plotted in log scale.

(f) Dual solution and primal reconstruction

The reduced dual problem $\min_\theta g(\theta)$ is solved using the L-BFGS-B algorithm. The optimal dual value obtained is

$$g(\theta^\star) = 2.270053591755101.$$

From $\theta^\star$ we reconstruct the Gibbs distribution

$$\tilde{p}_i = \frac{\exp(\theta^{\star\top}\phi_i)}{\sum_j \exp(\theta^{\star\top}\phi_j)}.$$

The reconstructed distribution agrees extremely closely with the primal solution:

$$\|p^\star - \tilde{p}\|_\infty = 1.9825705620812872 \times 10^{-6}.$$

The numerical duality gap is similarly negligible:

$$|H(p^\star) - g(\theta^\star)| = 3.712860241478211 \times 10^{-9}.$$

These results confirm numerical strong duality and verify that the Gibbs reconstruction matches the primal optimum to high precision.

Question 2: Binary Classification (Logistic Regression)

We consider the unregularized logistic loss

$$L(w) = \sum_{i=1}^{n} \log\big(1 + \exp(-y_i x_i^\top w)\big), \qquad y_i \in \{\pm 1\}.$$

for the following data

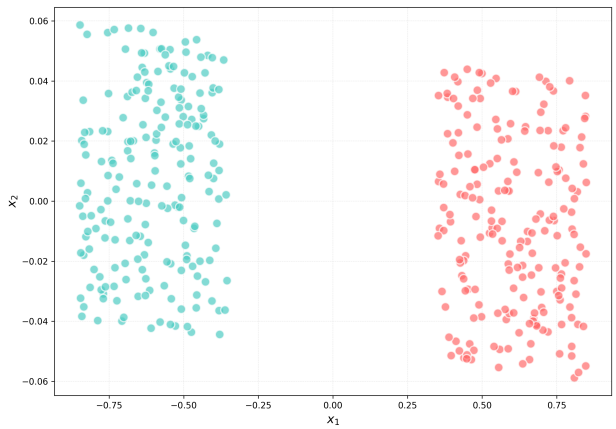


Figure 2: We can notice that the data is linearly separable.

(a) Probabilistic interpretation

The logistic model defines class probabilities

$$\Pr(y = +1 \mid x; w) = \sigma(x^\top w) = \frac{1}{1 + e^{-x^\top w}},$$

so minimizing the logistic loss corresponds to maximum likelihood estimation of a probabilistic classifier that separates the two classes by the sign of $x^\top w$.

(b) Numerical solver

We solve (1) using the L-BFGS-B algorithm (SciPy). At each iteration we record the objective value $L(w^k)$ and the weight norm $\|w^k\|_2$. The method converges in the sense that the gradient norm becomes small, but the iterates diverge in magnitude.

$$L(w^{\text{final}}) = 2.2248711 \times 10^{-5}, \qquad \|w^{\text{final}}\|_2 = 38.67787.$$

(c) Why no finite optimizer exists

If the data are linearly separable, then for any separating direction $\hat{w}$ with $y_i x_i^\top \hat{w} > 0$, the loss satisfies

$$L(t\hat{w}) = \sum_{i=1}^{n} \log\big(1 + \exp(-t\, y_i x_i^\top \hat{w})\big) \to 0 \quad \text{as } t \to \infty,$$

while $\|t\hat{w}\|_2 \to \infty$. Hence the infimum of the logistic loss is 0, but it is not attained at any finite w. This exactly matches our numerical behavior: the loss drops to nearly zero while the norm grows unbounded.

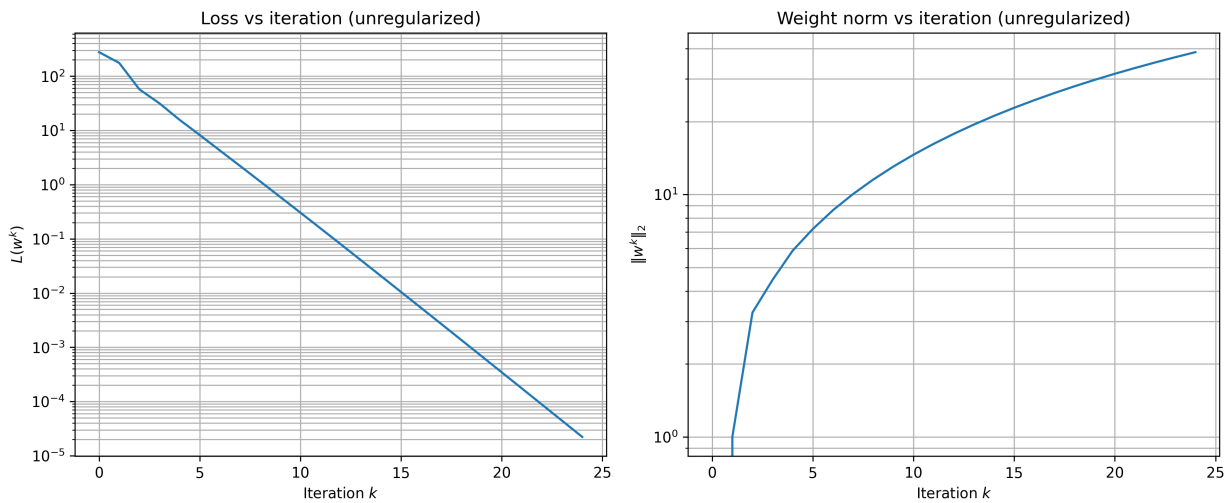


Figure 3: Unregularized logistic regression: (left) $k \mapsto L(w^k)$ (log scale) showing decay toward 0; (right) $k \mapsto \|w^k\|_2$ (log scale) showing divergence of the weight vector.

(d) Regularized formulation

To obtain a well-posed problem with a finite optimizer, we solve the ℓ_2-regularized logistic regression problem

$$L_\lambda(w) = \sum_{i=1}^{n} \log\big(1 + \exp(-y_i x_i^\top w)\big) + \frac{\lambda}{2}\|w\|_2^2, \qquad \lambda > 0,$$

which is strongly convex and has a unique minimizer.

Using $\lambda = 0.01$, L-BFGS-B converges to a stable solution:

$$L_\lambda(w^\star) = 1.36878939, \qquad \|w^\star\|_2 = 14.36292, \qquad \text{training accuracy} = 1.0.$$

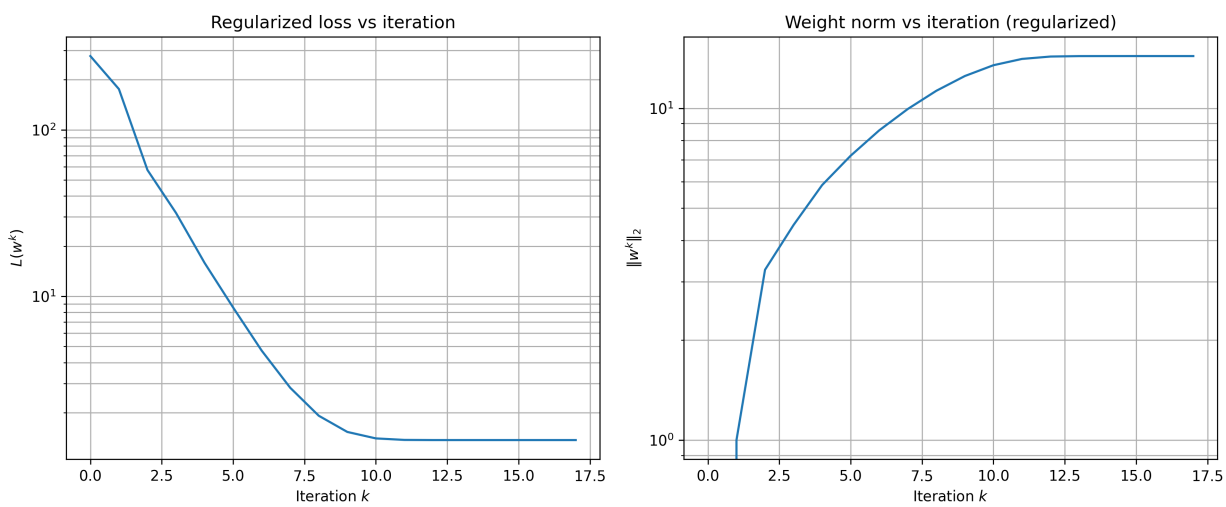


Figure 4: Regularized logistic regression ($\lambda = 0.01$): (left) $L_\lambda(w^k)$ converges smoothly; (right) $\|w^k\|_2$ remains bounded, demonstrating the stabilizing effect of regularization.

These results confirm that the unregularized problem is ill-posed on separable data, and that adding ℓ_2 regularization yields a well-defined and numerically stable optimizer.

Remark (behaviour of loss in log scale). For the unregularized logistic loss

$$L(w) = \sum_{i=1}^{n} \log\bigl(1 + \exp(-y_i x_i^\top w)\bigr),$$

linear separability implies the existence of u with $y_i x_i^\top u > 0$ for all i. Gradient-based methods generate iterates of the form

$$w_k = w_0 + t_k u + o(1), \qquad t_k \to +\infty,$$

and hence

$$L(w_k) = \sum_{i=1}^{n} e^{-y_i x_i^\top w_k}(1 + o(1)) = C\, e^{-t_k}(1 + o(1)).$$

Thus

$$\log L(w_k) = -t_k + \text{const},$$

so when t_k grows linearly with k (e.g. in GD or L-BFGS), the curve $k \mapsto L(w_k)$ is *linear in log scale.*

With ℓ_2–regularization,

$$L_\lambda(w) = L(w) + \frac{\lambda}{2}\|w\|_2^2, \qquad \lambda > 0,$$

the objective becomes λ–strongly convex and admits a finite minimizer $w^\star$. Standard convergence yields

$$\|w_k - w^\star\|_2 \le \rho^k \|w_0 - w^\star\|_2, \qquad 0 < \rho < 1,$$

and by quadratic growth,

$$L_\lambda(w_k) - L_\lambda(w^\star) = \Theta(\|w_k - w^\star\|_2^2) = \Theta(\rho^{2k}).$$

Hence

$$\log\bigl(L_\lambda(w_k) - L_\lambda(w^\star)\bigr) = 2k \log \rho + \text{const},$$

a curvature–dominated regime near $w^\star$, producing a *nonlinear* curve in log scale, in contrast to the unregularized case.

Question 3: Graph Coloring SDP

(a) Dual derivation

The primal problem is

$$\begin{aligned} &\min_{G,\rho} \rho \\ &\text{s.t. } G \succeq 0, \qquad G_{vv} = 1 \ \forall v, \qquad G_{uv} \le \rho \ \forall (u,v) \in E. \end{aligned}$$

Introduce multipliers λ_v for $G_{vv} = 1$, nonnegative $\alpha_{uv} \ge 0$ for $G_{uv} - \rho \le 0$, and a dual slack $S \succeq 0$. The Lagrangian is

$$\mathcal{L} = \rho + \sum_v \lambda_v (1 - G_{vv}) + \sum_{(u,v)\in E} \alpha_{uv}(G_{uv} - \rho) - \langle S, G \rangle.$$

The ρ–terms give $\rho(1 - \sum_e \alpha_{uv})$, so boundedness requires

$$\sum_{(u,v)\in E} \alpha_{uv} = 1.$$

Minimizing over $G \succeq 0$ yields the dual PSD constraint

$$A(\alpha) - \operatorname{diag}(\lambda) \succeq 0,$$

where $A(\alpha)$ has (u,v) and (v,u) entry α_{uv} for each edge. Thus the dual is

$$\boxed{\begin{aligned} &\max_{\lambda,\alpha} \sum_v \lambda_v \\ &\text{s.t. } A(\alpha) - \operatorname{diag}(\lambda) \succeq 0, \\ &\qquad \sum_{(u,v)\in E} \alpha_{uv} = 1, \qquad \alpha_{uv} \ge 0. \end{aligned}}$$

(b) Numerical results

<u>K_4(**Complete Graph with 4 nodes**):</u>

$$\rho^\star \approx -0.33333333, \qquad \sum_v \lambda_v^\star \approx -0.66666667, \qquad \text{gap} \approx 0.33333333.$$

<u>C_5(**Cycle Graph with 5 nodes**):</u>

$$\rho^\star \approx -0.80901723, \qquad \sum_v \lambda_v^\star \approx -1.61803699, \qquad \text{gap} \approx 0.80901976.$$

A key complementary–slackness identity is

$$\sum_v \lambda_v^\star = 2\rho^\star.$$

Indeed,

$$\langle A(\alpha^\star), G^\star \rangle = 2\sum_e \alpha_e^\star G_{uv}^\star = 2\rho^\star, \qquad \langle \operatorname{diag}(\lambda^\star), G^\star \rangle = \sum_v \lambda_v^\star,$$

and the PSD slackness condition forces their difference to be zero. Hence the duality gap (primal − dual) satisfies

$$\text{gap} = \rho^\star - 2\rho^\star = -\rho^\star > 0,$$

exactly matching the computed values for both graphs.

(c) Feasibility of 3-colorability

(i) Certificate

If a dual feasible point satisfies

$$\sum_v \lambda_v > -\tfrac{1}{2},$$

then by weak duality $\rho^\star \geq \sum_v \lambda_v > -1/2$. But the 3-color vector-coloring threshold requires $\rho \leq -1/2$. Thus no feasible (G, ρ) can satisfy the 3-color condition, proving infeasibility of the relaxation and hence non-3-colorability.

(ii) Interpretation for the graphs

<u>K$_4$:</u>

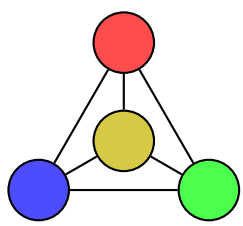
$$\rho^\star = -1/3 > -1/2.$$

The relaxation cannot reach the 3-color threshold; K_4 is not 3-colorable.

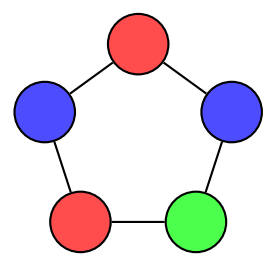
<u>C$_5$:</u>

$$\rho^\star \approx -0.809 < -1/2.$$

The relaxation is feasible at the 3-color threshold; C_5 is consistent with being 3-colorable.



(a) A proper 4-coloring of the complete graph K_4.



(b) A proper 3-coloring of the cycle graph C_5.

Figure 5: Side-by-side comparison of valid colorings for K_4 and C_5.